Assignment 1

1. Factorial with Edge Case Handling

Problem Description:

You are given a non-negative integer n. Write a recursive function to compute its factorial. The factorial of a number n is defined as:

- $n! = n \times (n-1) \times (n-2) \times ... \times 1$
- By definition, $0! = 1$

Your function must correctly handle the edge case when $n = 0$. Also consider what should happen if a negative number is passed.

Requirements:

- Use recursion only (no loops).
- Clearly define the base case.
- Ensure your function does not cause infinite recursion

2. Fibonacci Sequence with Recursive Definition

Problem Description:

The Fibonacci sequence is defined as:

- $F(0) = 0, F(1) = 1$
- $F(n) = F(n-1) + F(n-2)$ for $n \geq 2$

Write a recursive function that returns the nth Fibonacci number.

Example:
Input: $n = 5$
Output: 5

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3. Recursive String Reversal

Problem Description:

Given a string, write a recursive function to reverse it. You are not allowed to use loops or built-in reverse functions.

Hint:
Think about reducing the problem size by removing one character at a time and solving the smaller problem.

Example:
Input: `"recursion"`
Output: `"noisrucer"`

Requirements:

- Base case: when the string becomes empty or has one character.
- Recursive step: process the remaining substring.

4. Check if an Array is Sorted

Problem Description:
Given an integer array, determine whether it is sorted in ascending order using recursion.

Instead of checking the whole array at once, compare elements step by step recursively.

Approach Hint:

- Compare the current element with the next one.
- Then recursively check the rest of the array.

Example:
Input: `[2, 4, 6, 8]` → Output: `true`
Input: `[3, 5, 2, 9]` → Output: `false`

Requirements:

- Do not use loops.
- Use an index parameter to track position.

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